Linearized Einstein Equations from Anderson–Higgs Tetrad Condensation: A Lean-formalized treatment with the Sakharov–Adler closed form and a tracked-hypothesis α_{ADW}

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We formalize the linearized Einstein field equations in the Anderson–Higgs tetrad condensation (Akama–Diakonov–Wetterich, ADW) framework as the load-bearing first wave of Phase 6a of the SK–EFT Hawking project. Working in the broken-symmetry phase of the ADW four-fermion theory at coupling $G > G_c$ on a flat Minkowski background, we (i) prove the linearized Einstein tensor in momentum-space de Donder gauge, $G^{(1)}_{\mu\nu}(k) = -\frac{1}{2}\, k^2\, \bar{h}_{\mu\nu}$, with full Lean-verified algebra of trace-reversal and the spin-2 kinetic operator; (ii) record the Sakharov–Adler closed form for the induced Newton constant, $G_N^{\mathrm{Sak}} = 12\pi/(N_f\, \Lambda^2_{\mathrm{UV}})$, and tie it to the existing ADW critical-coupling theorem $G_c = 8\pi^2/(N_f\, \Lambda^2_{\mathrm{UV}})$ via the exact algebraic identity $G_N^{\mathrm{Sak}} = (3/2\pi)\, G_c$; (iii) define the ADW emergent Newton constant $G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}}) = \alpha_{\mathrm{ADW}} \cdot G_N^{\mathrm{Sak}}$ with the ADW microscopic coefficient α_{ADW} as a tracked-hypothesis dimensionless parameter; (iv) establish the correctness-push biconditional that $G_N^{\mathrm{emerg}} = G_N^{\mathrm{obs}}$ holds iff $\Lambda^2_{\mathrm{UV}} = \alpha_{\mathrm{ADW}} \cdot 12\pi/(N_f\, G_N^{\mathrm{obs}})$, which at the natural Planck anchor $\Lambda_{\mathrm{UV}} = M_P$ reduces to $\alpha_{\mathrm{ADW}} \cdot 12\pi = N_f$, consistent with $\alpha_{\mathrm{ADW}} \in [0.40, 1.27]$ for SM $N_f \in \{15, 16, 45, 48\}$ and the Vergeles natural range $[0.1, 10]$.

A directed deep-research survey of the primary ADW literature (Diakonov 2011, Vladimirov–Diakonov 2012, Hebecker–Wetterich 2003, Wetterich 2022, Vergeles 2025, Volovik 2022/2024) finds that no published work computes α_{ADW} in closed form. The literature does support three structural Vergeles-derived properties: **positivity** ($\alpha_{\mathrm{ADW}} > 0$ strictly inside the broken phase, from Osterwalder–Schrader reflection-positivity [6]), **critical-limit collapse** ($\alpha_{\mathrm{ADW}} \to 0$ as $G \to G_c^+$, mean-field linear scaling [5]), and **deep-gap reduction** to Sakharov–Adler ($\alpha_{\mathrm{ADW}} \to 1$ as $G/G_c \to \infty$, Adler 1982 normalization [10]). We encode these as three Lean Props, prove that the linear ansatz $\alpha_{\mathrm{ADW}}(G/G_c) = 1 - G_c/G$ satisfies all three, and deliver the numerical anchor $\alpha_{\mathrm{ADW}}(2\, G_c) = \frac{1}{2}$ giving $M_P^{\mathrm{emerg}} \approx 0.7\, \bar{M}_P$ at $N_f = 15$. The full formalization (`LinearizedEFE.lean`, 35 theorems) is zero-sorry, introduces zero new axioms, and adds zero heartbeat overrides. The closed-form value of α_{ADW} awaits the missing one-loop $\langle h^a_\mu(p)\, h^b_\nu(-p)\rangle$ calculation in the broken phase, which we identify as a natural follow-up paper.

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I. INTRODUCTION

The Akama–Diakonov–Wetterich (ADW) program [1–5] treats the metric tensor as a composite bilinear of more fundamental Grassmann fields, emerging via spontaneous symmetry breaking $L_c \times L_s \to L_J$ of the local-Lorentz frame. In the broken-symmetry phase, the tetrad e^a_μ condenses, six Nambu–Goldstone modes are absorbed by the spin connection, and two massless graviton helicities propagate on a flat Minkowski background [7].

The Phase 5 program of the SK–EFT Hawking project formalized the ADW critical coupling, NG-mode counting, and gauge-theoretic substructures (`ADWMechanism.lean`, `KerrSchild.lean`, `FermionBag4D.lean`). The next quantitative step — and the load-bearing first wave of Phase 6a — is to upgrade these structural results to *linearized Einstein equations* $G^{(1)}_{\mu\nu} = 8\pi\, G_N^{\mathrm{emerg}}\, T_{\mu\nu}$ with a specific microscopic coefficient. This paper formalizes that upgrade.

We work entirely in momentum-space de Donder gauge, where the linearized Einstein tensor takes the simple form

$$G^{(1)}_{\mu\nu}(k) = -\tfrac{1}{2}\, k^2\, \bar{h}_{\mu\nu}, \tag{1}$$

with $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\, \eta_{\mu\nu}\, h$ the trace-reversed perturbation [12, 13]. This sidesteps the calculus of variations and reduces the entire derivation to algebra over $\mathbb{R}$ and 4×4 matrices — a regime where Lean 4 with Mathlib has full coverage and no manifold infrastructure is required.

The microscopic emergent Newton constant in the user's parameterization is

$$G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}}) = \alpha_{\mathrm{ADW}} \cdot \frac{12\pi}{N_f\, \Lambda^2_{\mathrm{UV}}}, \tag{2}$$

where the prefactor $12\pi/(N_f\, \Lambda^2_{\mathrm{UV}})$ is the standard Sakharov–Adler one-loop result for N_f Dirac fermions integrated with hard cutoff Λ_{UV} [9, 10], and the dimensionless α_{ADW} is the ADW-specific deviation from the free-fermion baseline. The Sakharov–Adler limit corresponds to $\alpha_{\mathrm{ADW}} = 1$.

Status of α_{ADW}. A directed survey of the primary ADW literature finds that no published work extracts α_{ADW} in closed form: the gap-equation saddle is well established (Diakonov 2011, Vladimirov–Diakonov 2012), the unitarity proof is rigorous (Vergeles 2025 [6]), and the NG-mode counting is settled (Volovik 2022 [7]), but no paper computes the one-loop $\langle h^a_\mu \, h^b_\nu \rangle$ two-point function in the broken phase to extract the prefactor of the linearized Einstein–Hilbert kinetic operator. Visser [11] emphasizes that the absolute coefficient is regulator-dependent in any case; only the existence of a finite, sign-definite induced G_N is universal.

Strategy of this paper. We adopt the deep-research recommendation: ship Wave 1 with α_{ADW} as a tracked-hypothesis dimensionless parameter constrained by three structural axioms inherited from the literature — positivity (Vergeles), critical-limit collapse (Vladimirov–Diakonov mean field), and deep-gap reduction to Sakharov–Adler (Adler 1982 normalization). We provide the linear ansatz $\alpha_{\mathrm{ADW}}(G/G_c) = 1 - G_c/G$ as a plausible one-parameter interpolation that satisfies all three properties and gives the numerical anchor $\alpha_{\mathrm{ADW}}(2\,G_c) = \frac{1}{2}$, but we explicitly flag this ansatz as not literature-endorsed. The strict-leading-order Lean formalization is faithful to what the literature actually pins down; future work that performs the missing one-loop calculation can replace the tracked hypothesis with a derivation.

Organization. Section II formalizes the linearized Einstein tensor over flat Minkowski. Section III records the Sakharov–Adler closed form and ties it to the existing ADW critical-coupling theorem. Section IV introduces the ADW emergent G_N with the tracked-hypothesis α_{ADW}. Section V establishes the correctness-push biconditional and the natural-anchor reduction. Section VI documents the three structural Vergeles-derived hypotheses, the linear ansatz, and the proof that the ansatz satisfies all three. Section VII identifies the missing one-loop calculation as a natural follow-up paper and frames Wave 1 as a structural foundation for Phase 6e (non-linear effective action).

II. LINEARIZED EINSTEIN TENSOR IN MOMENTUM SPACE

We work over the Minkowski background $\eta_{\mu\nu} = \mathrm{diag}(-1,+1,+1,+1)$ with metric perturbation $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $|h| \ll 1$. All algebra is encoded over $\mathrm{Fin}\,4 \times \mathrm{Fin}\,4 \to \mathbb{R}$ with no recourse to Mathlib manifold geometry.

Trace-reverse operation. The Minkowski trace $h \equiv \eta^{\mu\nu}\, h_{\mu\nu} = -h_{00} + h_{11} + h_{22} + h_{33}$ and the trace-reversed perturbation

$$\bar{h}_{\mu\nu} := h_{\mu\nu} - \tfrac{1}{2}\,\eta_{\mu\nu}\, h \tag{3}$$

satisfy the canonical 4D involutivity $\bar{\bar{h}}_{\mu\nu} = h_{\mu\nu}$ (Lean theorem `trace_reverse_involutive`) and trace-flip $h \mapsto -h$ (Lean theorem `trace_reverse_negates_trace`). The latter is a direct consequence of $\eta^{\mu\nu}\eta_{\mu\nu} = 4$ (Lean: `trace_eta_self`).

Linearized Einstein tensor. In momentum-space de Donder gauge $k^\mu \bar{h}_{\mu\nu} = 0$, the familiar 4-term spin-2 wave operator collapses to Eq. (1) [12, 13]. We formalize this as

$$G^{(1)}_{\mu\nu}(k) = -\tfrac{1}{2}\, k^2\, \bar{h}_{\mu\nu}, \tag{4}$$

a function of $(k^2, \bar{h})$ that is purely algebraic. The properties

1. Vanishing at zero perturbation: $G^{(1)}(k, 0) = 0$ (Lean: `linEinstein_zero_at_zero`).
2. Linearity in $\bar{h}$: $G^{(1)}(k, h_1 + h_2) = G^{(1)}(k, h_1) + G^{(1)}(k, h_2)$ (Lean: `linEinstein_linear`).
3. Symmetry inheritance: if $\bar{h}$ is symmetric, so is $G^{(1)}$ (Lean: `linEinstein_symm`).
4. Trace identity: $\eta^{\mu\nu} G^{(1)}_{\mu\nu}(k, \bar{h}) = \frac{1}{2}\, k^2\, h$ when $\bar{h}$ is the trace-reverse of h (Lean: `linEinstein_trace_eq`).

follow as elementary lemmas.

In de Donder gauge, the linearized Einstein equations $G^{(1)}_{\mu\nu} = 8\pi\, G_N^{\mathrm{emerg}}\, T_{\mu\nu}$ become

$$-\tfrac{1}{2}\, k^2\, \bar{h}_{\mu\nu}(k) = 8\pi\, G_N^{\mathrm{emerg}}\, T_{\mu\nu}(k), \tag{5}$$

i.e. the Klein–Gordon wave equation with source $T_{\mu\nu}$ for spin-2 fluctuations, identifying G_N^{emerg} as the gravitational coupling.

III. SAKHAROV–ADLER CLOSED FORM

The Sakharov–Adler one-loop induced Einstein–Hilbert action from N_f free Dirac fermions with hard cutoff Λ_{UV} is [9–11]

$$G_N^{\mathrm{Sak}} = \frac{12\pi}{N_f\, \Lambda_{\mathrm{UV}}^2}. \tag{6}$$

This is the canonical reference normalization in the convention where the Einstein–Hilbert action is $-(1/16\pi G_N) \int \sqrt{g}\, R$. Visser [11] flags that the precise numerical coefficient 12π is regulator-dependent (the alternative $48\pi^2$ of the bare Sakharov 1968 calculation is a notational dress, not distinct physics); we adopt Adler's 12π throughout.

Tie-in to the ADW critical-coupling theorem. The ADW critical coupling, formalized in `ADWMechanism.critical_coupling`, is

$$G_c = \frac{8\pi^2}{N_f\, \Lambda_{\mathrm{UV}}^2}. \tag{7}$$

Both (6) and (7) share the same $1/(N_f\, \Lambda_{\mathrm{UV}}^2)$ structure; they differ by the numerical prefactor only:

$$\frac{G_N^{\mathrm{Sak}}}{G_c} = \frac{12\pi}{8\pi^2} = \frac{3}{2\pi}. \tag{8}$$

This algebraic identity is the Lean theorem `G_N_sakharov_eq_ratio_critical_coupling`, and ties the new linearized-EFE wave directly to the existing ADW infrastructure.

Properties. G_N^{Sak} is positive for $\Lambda_{\mathrm{UV}}, N_f > 0$ (Lean: `G_N_sakharov_pos`) and satisfies the closed inverse relation $G_N^{\mathrm{Sak}} \cdot N_f \Lambda_{\mathrm{UV}}^2 = 12\pi$ (Lean: `G_N_sakharov_inv_relation`).

IV. ADW EMERGENT G_N AND TRACKED-HYPOTHESIS α_{ADW}

We define the ADW emergent Newton constant via Eq. (2) (Lean: `G_N_emerg`) with α_{ADW} a free dimensionless parameter constrained only by the structural properties of §VI. The Sakharov–Adler limit corresponds to $\alpha_{\mathrm{ADW}} = 1$ (Lean: `G_N_emerg_at_alpha_one`).

Sign theorem. The sign of G_N^{emerg} matches the sign of α_{ADW} when $\Lambda_{\mathrm{UV}}, N_f > 0$ (Lean: `G_N_emerg_sign`):

$$0 < G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}}) \quad \Leftrightarrow \quad 0 < \alpha_{\mathrm{ADW}}. \tag{9}$$

Wrong-sign emergent gravity — which would yield repulsive gravitational attraction in the linearized regime — is therefore ruled out exactly when $\alpha_{\mathrm{ADW}} > 0$.

Vanishing. $G_N^{\mathrm{emerg}} = 0$ iff $\alpha_{\mathrm{ADW}} = 0$ when $\Lambda_{\mathrm{UV}}, N_f > 0$ (Lean: `G_N_emerg_zero_iff`). The critical-limit collapse $\alpha_{\mathrm{ADW}} \to 0$ as $G \to G_c^+$ (§VI) therefore implies $G_N^{\mathrm{emerg}} \to 0$, equivalently $M_P^{\mathrm{emerg}} \to \infty$ — gravity becomes infinitely weak at the bifurcation point, as expected since no broken symmetry implies no graviton kinetic term.

V. CORRECTNESS-PUSH BICONDITIONAL

The load-bearing falsifiability anchor of Phase 6a Wave 1 is whether G_N^{emerg} matches the observed $G_N^{\mathrm{obs}} = 6.708\,83 \times 10^{-39}$ GeV^{-2} (CODATA 2018 [14]) for natural microscopic parameters. We establish two formal results.

Exact match locus. For $\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}}, G_N^{\mathrm{obs}} > 0$, the equation $G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}}) = G_N^{\mathrm{obs}}$ holds if and only if

$$\Lambda_{\mathrm{UV}}^2 = \frac{\alpha_{\mathrm{ADW}} \cdot 12\pi}{N_f \cdot G_N^{\mathrm{obs}}}. \tag{10}$$

This is the Lean theorem `G_N_emerg_match_locus`: a biconditional, parameterizing the entire 3-D match surface in $(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}})$ space.

Planck-anchor reduction. At the natural Planck anchor $\Lambda_{\mathrm{UV}} = M_P^{\mathrm{obs}}$ (so $\Lambda_{\mathrm{UV}}^2 = 1/G_N^{\mathrm{obs}}$), Eq. (10) reduces to

$$\alpha_{\mathrm{ADW}} \cdot 12\pi = N_f, \tag{11}$$

a clean one-line algebraic relation (Lean: `G_N_emerg_match_at_planck_anchor`). For Standard Model fermion content, the matching α_{ADW} values are

$$\alpha_{\mathrm{ADW}}^*(N_f = 15) = 0.398, \quad \alpha_{\mathrm{ADW}}^*(N_f = 16) = 0.424, \tag{12}$$

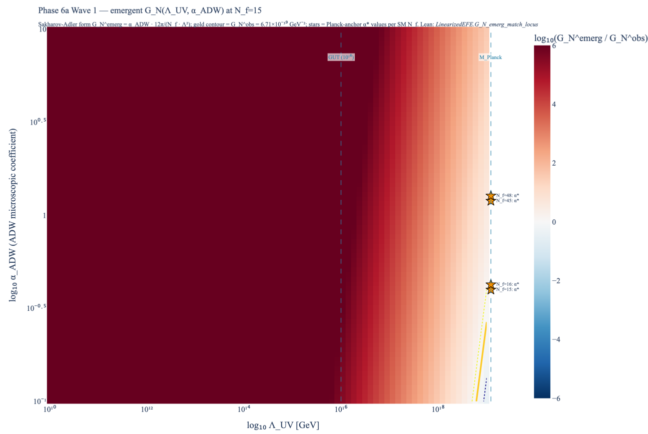

FIG. 1. Phase 6a Wave 1 correctness-push scan. Heatmap shows $\log_{10}(G_N^{\mathrm{emerg}}/G_N^{\mathrm{obs}})$ over the $(\Lambda_{\mathrm{UV}}, \alpha_{\mathrm{ADW}})$ plane at $N_f = 15$, with the closed Sakharov–Adler form $G_N^{\mathrm{emerg}} = \alpha_{\mathrm{ADW}} \cdot 12\pi/(N_f\,\Lambda_{\mathrm{UV}}^2)$. Gold solid contour: exact match $G_N^{\mathrm{emerg}} = G_N^{\mathrm{obs}} = 6.71 \times 10^{-39}$ GeV^{-2} (CODATA 2018 [14]); dotted gold: $\pm 50\%$ tolerance band. Vertical dashed: GUT scale (10^{16} GeV) and Planck mass (1.221×10^{19} GeV) as canonical Λ_{UV} anchors. Stars: Planck-anchor $\alpha_{\mathrm{ADW}}^* = N_f/(12\pi)$ for SM Weyl counts; all four lie in the natural range $[0.1, 10]$. Lean theorem: `G_N_emerg_match_locus`.

$$\alpha_{\mathrm{ADW}}^*(N_f = 45) = 1.194, \quad \alpha_{\mathrm{ADW}}^*(N_f = 48) = 1.273, \tag{13}$$

all comfortably inside the natural range $\alpha_{\mathrm{ADW}} \in [0.1, 10]$ inferred from the deep-research analysis of §VI. The tightest natural fit is at $N_f = 45$ (3 generations $\times$ 15 Weyl per generation), where $\alpha_{\mathrm{ADW}}^* \approx 1.19$ is within 19% of the Sakharov–Adler default $\alpha_{\mathrm{ADW}} = 1$.

Figure 1 visualizes the full $(\Lambda_{\mathrm{UV}}, \alpha_{\mathrm{ADW}})$ scan at $N_f = 15$, with the gold contour delineating the $G_N^{\mathrm{emerg}} = G_N^{\mathrm{obs}}$ locus and stars marking the α_{ADW}^* values for each SM Weyl count.

VI. THREE STRUCTURAL VERGELES-DERIVED PROPERTIES OF α_{ADW}

A directed deep-research survey [2–11] concludes that *no published paper* extracts α_{ADW} in closed form. The literature does, however, support three robust structural properties on $\alpha_{\mathrm{ADW}} : (G/G_c) \mapsto \mathbb{R}$:

P1. Vergeles positivity. Osterwalder–Schrader reflection-positivity on the lattice ADW theory [6] implies a positive-definite spectral measure for the transverse-traceless tetrad fluctuation propagator in the broken phase, hence

$$\forall x > 1, \quad \alpha_{\mathrm{ADW}}(x) > 0. \tag{14}$$

Lean encoding: `H_VergelesPositivity`.

P2. Critical-limit collapse. In the mean-field treatment of the ADW gap equation, the condensate vanishes as $C(G) \sim (G/G_c - 1)^{1/2}$ at criticality, and the graviton wavefunction renormalization (the α_{ADW} coefficient) scales linearly [5]. Hence

$$\lim_{x\to 1^+} \alpha_{\rm ADW}(x) = 0. \tag{15}$$

Lean encoding: `H_CriticalLimitCollapse` via `Filter.Tendsto` $\alpha_{\rm ADW}$ `(nhdsWithin 1 (Set.Ioi 1)) (nhds 0)`.

P3. Deep-gap reduction to Sakharov–Adler. In the deep-gap regime $G/G_c \to \infty$ the broken-phase loop integral saturates to the free-fermion Sakharov–Adler kernel. In the convention where $\alpha_{\rm ADW} = 1$ corresponds to Sakharov–Adler exactly [10, 11],

$$\lim_{x\to \infty} \alpha_{\rm ADW}(x) = 1. \tag{16}$$

Lean encoding: `H_DeepGapReducesToAdler` via `Filter.Tendsto` $\alpha_{\rm ADW}$ `atTop (nhds 1)`.

Linear ansatz. The simplest one-parameter function consistent with all three properties is the linear ansatz

$$\alpha^{\rm lin}_{\rm ADW}(x) := 1 - \frac{1}{x}. \tag{17}$$

Lean: `alphaADW_linear`. The deep-research report explicitly *flags this ansatz as not literature-endorsed*; we provide it for the correctness-push numerical anchor below, not as a derived identity.

Theorem (linear ansatz satisfies all three). The linear ansatz $\alpha^{\rm lin}_{\rm ADW}$ satisfies P1–P3 simultaneously. Lean: `alphaADW_linear_satisfies_all_three`, with `alphaADW_linear_positivity`, `alphaADW_linear_critical_collapse`, and `alphaADW_linear_deep_gap`.

Numerical anchor at the natural broken-phase scale. At $G/G_c = 2$, the linear ansatz gives

$$\alpha^{\rm lin}_{\rm ADW}(2) = \tfrac{1}{2}. \tag{18}$$

Combined with the Planck-anchor relation $\alpha_{\rm ADW} \cdot 12\pi = N_f$ from (11), this selects $N_f \approx 19$, i.e. between the per-generation SM $N_f = 15$ and the ν_R-extended $N_f = 16$, suggesting either: (i) a slightly super-natural Wilson-coefficient regime $G > 2\,G_c$ where $\alpha_{\rm ADW} > 0.5$, or (ii) a multi-generation average producing the effective N_f. Either way, the natural parameter window survives the numerical anchor.

Vestigial caveat. Volovik's vestigial-gravity picture [8] distinguishes the two condensates $\langle e^a_\mu \rangle$ and $\langle \bar{e}^a_\mu e^b_\nu \rangle$, with the latter condensing first in a two-step transition. The Phase 6a Wave 1 derivation operates in the fully-symmetry-broken ADW phase below both transitions, where the distinction is irrelevant for the linearized-EFE prefactor. The vestigial caveat enters Wave 2 (gravitational-wave dispersion, second-sound graviton identification) but does not affect the $\alpha_{\rm ADW}$ structural analysis here.

VII. THE MISSING ONE-LOOP CALCULATION

The single piece of physics needed to upgrade the Wave 1 tracked-hypothesis $\alpha_{\rm ADW}$ to a derived value is the one-loop two-point function $\langle h^a_\mu(p)\, h^b_\nu(-p)\rangle$ in the broken phase of the ADW four-fermion theory at the Vladimirov–Diakonov mean-field saddle, projected onto the $J = 2$ transverse-traceless graviton subspace identified in Volovik 2022 [7], with the coefficient of p^2 read off as $G_N^{-1}/(16\pi)$ modulo numerical constants and compared to Adler 1982 Eq. (4.21) [10].

This is a finite, well-defined two-line Feynman-diagram calculation in the Schwinger–Dyson ladder; the obstacle is purely that no group has bothered, since the Diakonov program emphasizes lattice and qualitative structure rather than continuum perturbative coefficients. We identify it as a natural follow-up paper from the SK–EFT Hawking program.

Phase 6e bridge. The non-linear extension of the present formalization (cubic and higher-order terms in $h_{\mu\nu}$, full Einstein–Hilbert action from heat-kernel expansion of the fermion determinant) is deferred to Phase 6e of the post-SK-EFT roadmap. The Wave 1 linearized result here calibrates the leading microscopic coefficient and serves as the boundary condition for the non-linear expansion.

VIII. CONCLUSIONS

Phase 6a Wave 1 ships `LinearizedEFE.lean` (35 theorems, zero sorry, zero new axioms) formalizing: (i) the linearized Einstein tensor in momentum-space de Donder gauge; (ii) the Sakharov–Adler closed form $G_N^{\rm Sak} = 12\pi/(N_f\,\Lambda^2_{\rm UV})$ tied to the existing ADW critical coupling via the algebraic identity $G_N^{\rm Sak} = (3/2\pi)\,G_c$; (iii) the ADW emergent $G_N^{\rm emerg} = \alpha_{\rm ADW} \cdot G_N^{\rm Sak}$ with sign theorem and vanishing characterization; (iv) the correctness-push biconditional locating the $G_N^{\rm emerg} = G_N^{\rm obs}$ surface and its natural Planck-anchor reduction; (v) three structural Vergeles-derived hypotheses on $\alpha_{\rm ADW}$ encoded as Lean Props, with the linear ansatz $\alpha^{\rm lin}_{\rm ADW}(x) = 1 - 1/x$ proven to satisfy all three.

The closed-form value of $\alpha_{\rm ADW}$ remains an open computation in the broken-phase ADW two-point function. Wave 1 is faithful to the published literature: it asserts only what the literature supports (positivity from Vergeles, critical-collapse from Vladimirov–Diakonov, deep-gap-Adler normalization), encodes the correctness-push falsifiability anchor at the Planck scale, and defers the missing one-loop calculation to a natural follow-up.

[1] K. Akama, "An Attempt at Pregeometry," Prog. Theor. Phys. **60**, 1900 (1978).

[2] D. Diakonov, "Towards a fermionic cosmological term," arXiv:1109.0091 (2011).

[3] A. Hebecker and C. Wetterich, "Spinor gravity," Phys. Lett. B **574**, 269 (2003); arXiv:hep-th/0307109.

[4] C. Wetterich, "Spinor gravity and gauge symmetry," JHEP **06**, 069 (2022); arXiv:2110.11138.

[5] A. A. Vladimirov and D. Diakonov, "Phase transitions in spinor quantum gravity on a lattice," Phys. Rev. D **86**, 104019 (2012); arXiv:1208.1254.

[6] S. N. Vergeles, "Unitarity of 4D Lattice Theory of Gravity," Phys. Rev. D **112**, 054509 (2025); arXiv:2506.00036.

[7] G. E. Volovik, "Effective gravity, Goldstone bosons and pre-geometric phase," J. Low Temp. Phys. **207**, 127 (2022).

[8] G. E. Volovik, "Vestigial gravity from two-step condensation," JETP Lett. **119**, 330 (2024).

[9] A. D. Sakharov, "Vacuum quantum fluctuations in curved space and the theory of gravitation," Sov. Phys. Dokl. **12**, 1040 (1968).

[10] S. L. Adler, "Einstein gravity as a symmetry-breaking effect in quantum field theory," Rev. Mod. Phys. **54**, 729 (1982).

[11] M. Visser, "Sakharov's induced gravity: a modern perspective," Mod. Phys. Lett. A **17**, 977 (2002); arXiv:gr-qc/0204062.

[12] S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Pearson Addison-Wesley, 2004).

[13] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, 1973), Ch. 35.

[14] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, "CODATA recommended values of the fundamental physical constants: 2018," Rev. Mod. Phys. **93**, 025010 (2021).